STAC WARS · PS 4

ORBITAL FUEL

Making Asteroid Mining Economically Viable

SUBPART A · ENGINEERING

How We Reach, Mine & Operate

Reaching the Asteroid

Solar-electric propulsion for efficient transfer. Cameras and nav sensors map the surface on arrival. Slow approach, then mechanical anchoring.

Extracting & Processing

Water: Drill ice → heat in sealed chamber → capture vapour
Iron & Nickel: Magnetic separation → transport to refinery
PGMs: Survey and assay first → extract only if justified

Processing Strategy

Process enough at the asteroid to cut transported mass. Avoid expensive operations that cost more than they save.

Energy & Power

Large solar arrays sized for the asteroid belt. Batteries for short-term high-power demand. Mining scheduled to match available power.

Communication & Autonomy

High-gain antenna for data and status. Backup low-gain antenna and onboard storage. Semi-autonomous: routine tasks onboard, critical decisions sent to Earth.

Failure Handling

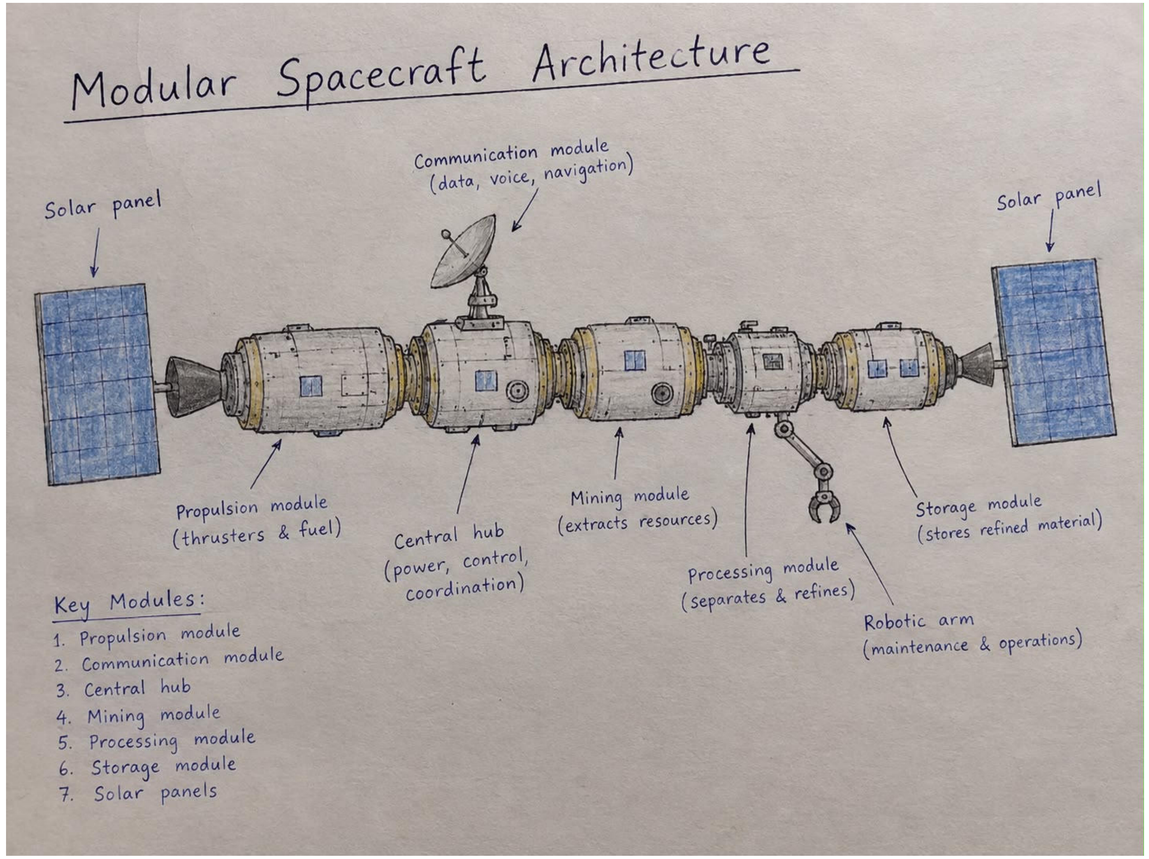
Drill jam: Stop and attempt controlled recovery
Anchor slip: Pause, reassess, reposition
Power loss: Batteries sustain essentials, non-critical off
Comms loss: Continue safe pre-programmed ops
Component fail: Modular isolation, critical redundancy

SUBPART A · SYSTEM DIAGRAM

Modular Spacecraft Architecture

7 Modular Subsystems

- → **Propulsion:** Solar-electric for efficient long-distance transfer
- → **Solar Power:** Large arrays + batteries for peak demand
- → **Mining:** Robotic drills, scoops, material handling
- → **Anchoring:** SmartGrip with force-feedback sensors
- → **Processing:** Heating chamber + metal concentration
- → **Storage:** Sealed water tanks + solid containers
- → **Comms & Nav:** Antennas, cameras, star trackers, autonomy software



BONUS · SUBPART A

SmartGrip Anchoring System

The Bottleneck

Asteroids have extremely weak gravity. Drilling force pushes the spacecraft away from the surface. Anchors can slip, causing failed attempts and equipment damage.

Our Solution

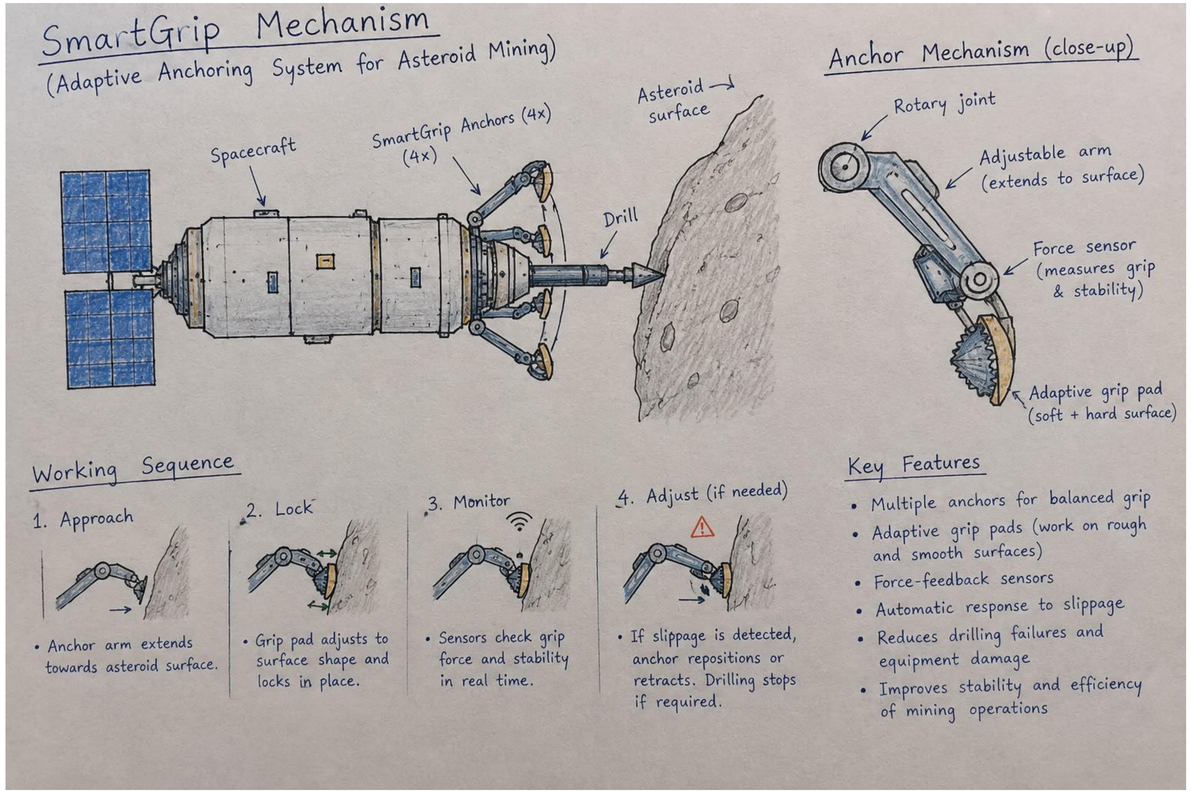
SmartGrip uses multiple mechanical anchors with force-feedback sensors distributed across several anchoring points.

How It Works

1. Distributes forces across multiple anchor points
2. Continuously monitors stability during drilling
3. If slipping detected → auto-stop drilling
4. Adjusts anchoring configuration automatically
5. Re-stabilises before restarting operations

Result

Fewer failed drilling attempts, less equipment damage, reduced



SUBPART B — COST

Cost Optimisation

Three Major Cost Drivers

1. Launch & Transportation

Rideshare launches when trajectory is compatible. Optimised asteroid target and launch window selection. Solar-electric propulsion to cut propellant mass. Spacecraft carries only equipment needed for the selected target. Rideshare evaluated against additional transfer cost.

2. Mining & Processing Equipment

Modular multi-task robotic tools. Survey and analyse asteroid before committing to extraction. Concentrate valuable material instead of transporting waste. No full refinery on the spacecraft. Autonomous monitoring to detect problems early and reduce unnecessary operations.

3. Transport & Operations

Separate useful fractions at the asteroid where justified. Transport only recovered water and concentrated metal. Autonomous fault detection and safe-mode procedures. Share downstream refining and manufacturing facilities with other missions. Processing amount depends on whether savings exceed added cost.

Bonus — Conventional vs Orbital Fuel

Key Assumptions & Viability

Key Assumptions

Key Assumptions

- → Suitable asteroid identified and resource content verified before investment
- → Solar-electric propulsion and modular mining equipment can be developed
- → Concentration at asteroid justifies added processing cost
- → Future orbital customers and processing facilities will exist
- → Resource quality and delivery requirements confirmed before commercial operations

Viability Test

Profit = Revenue − Total Costs
Total cost includes launch, transportation, mining, processing, storage, refining, delivery and maintenance. Proceed only when selling price exceeds complete delivery cost with safety margin.

Bonus — Conventional vs Orbital Fuel

Conventional: Full refinery on spacecraft, transport raw unprocessed ore, single-destination delivery, high propellant mass with chemical rockets

Orbital Fuel: Concentrate at asteroid and ship less waste, modular

SUBPART C · BUSINESS

Resources, Customers & Revenue

What We Sell

Water ice: Purified for life support or propellant production
Iron: Concentrated for space-based manufacturing
Nickel: Concentrated for alloys and components
PGMs: High-value metals, subject to verified economics

Our Customers

Orbital depots & stations: Water for life support and storage
Spacecraft companies: Metal feedstock for construction
Lunar missions: Water and materials for operations
PGM buyers: Refined metals if economically competitive

Where We Deliver

Water: Sell in space to orbital depots
Iron & Nickel: Shared space-based manufacturing facility
PGMs: Return to Earth only if value justifies cost
Rule: Best economic return per resource, not one destination for all

3 Revenue Streams

1. Water Sales

Sell purified water to orbital depots and space customers. Revenue depends on quantity, purity and agreed price.

2. Metal Feedstock & Components

Supply concentrated material to shared processing facility. Long-term: sell useful products, not raw asteroid material.

3. Platinum-Group Metals

Sell verified PGM concentrate or refined metals when assay results and delivery costs support a profitable transaction.

SUBPART C · RISKS & ROADMAP

Business Risks & Staged Roadmap

5 Major Risks & Mitigations

1. Uncertain Resource Content

Remote surveys, sample analysis and prospecting missions before full-scale investment.

2. High Transport & Processing Costs

Optimise trajectories, concentrate useful material, compare total delivered costs with alternatives.

3. Insufficient Customer Demand

Identify customers early, seek purchase commitments, scale to demonstrated demand.

4. Equipment Failure

Modular equipment, autonomous fault detection, safe-mode procedures, critical redundancy.

5. Immature Space Infrastructure

Begin with limited demo mission, expand only as infrastructure and customer base develop.

6-Stage Profitability Roadmap

1. **Prospect**
   Identify asteroid and verify resource composition
2. **Demonstrate**
   Test small-scale extraction, anchoring, water recovery
3. **Secure Customers**
   Confirm specs, demand and expected selling prices
4. **Deliver Test Batch**
   Demonstrate transport and delivery of recovered material
5. **Validate Economics**
   Calculate all costs vs actual customer willingness to pay
6. **Scale Operations**
   Expand only when process is reliable and revenue exceeds lifecycle costs

Thank You

Mine selectively · Transport less waste · Operate autonomously · Sell where value is greatest

ORBITAL FUEL — Turning asteroid resources into useful products for the future of space